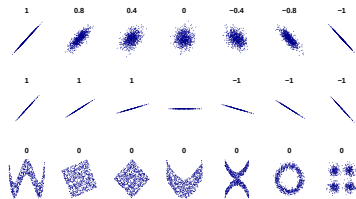
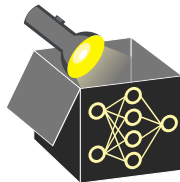


Interpretable Machine Learning

Intro to IML

Correlation and Dependencies



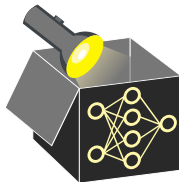
Learning goals

- Pearson correlation
- Coefficient of determination R^2
- Mutual information
- Correlation vs. dependence

PEARSON'S CORRELATION COEFFICIENT ρ

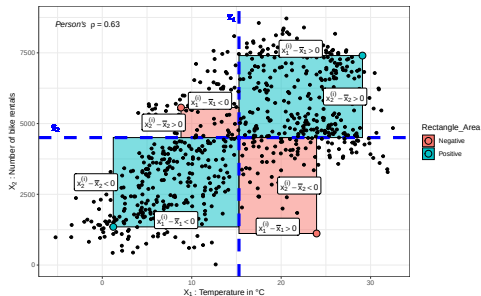
Correlation often refers to Pearson's correlation (captures only **linear relationship**)

$$\rho(X_1, X_2) = \frac{\sum_{i=1}^n (x_1^{(i)} - \bar{x}_1)(x_2^{(i)} - \bar{x}_2)}{\sqrt{\sum_{i=1}^n (x_1^{(i)} - \bar{x}_1)^2 \sum_{i=1}^n (x_2^{(i)} - \bar{x}_2)^2}} \in [-1, 1]$$



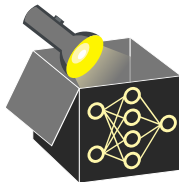
Geometric interpretation of ρ :

- Numerator is sum of rectangle's area with width $x_1^{(i)} - \bar{x}_1$ and height $x_2^{(i)} - \bar{x}_2$
- Areas enter numerator with positive (+) or negative (-) sign, depending on position
- Denominator scales the sum into the range $[-1, 1]$



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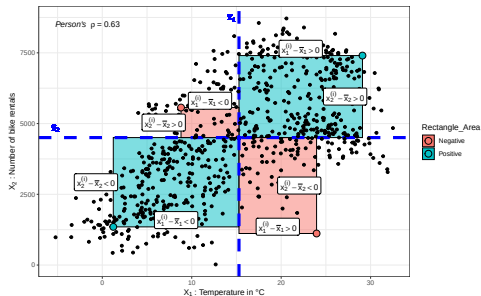
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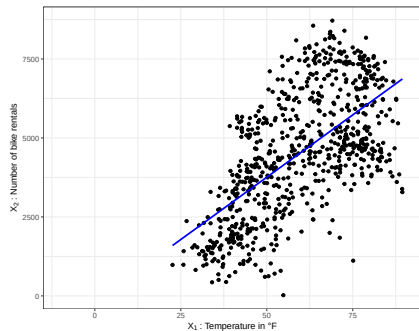
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- Areas enter numerator with positive (+) or negative (-) sign, depending on position
- Denominator scales the sum into the range $[-1, 1]$



- $\rho > 0$ if **positive areas** dominate **negative areas**
 $\rightsquigarrow X_1, X_2$ positive correlated
- $\rho < 0$ if **negative areas** dominate **positive areas**
 $\rightsquigarrow X_1, X_2$ negative correlated
- $\rho = 0$ if area of rectangles cancels out $\rightsquigarrow X_1, X_2$ linearly uncorrelated

COEFFICIENT OF DETERMINATION R^2

Another method to evaluate **linear dependency** between features is R^2

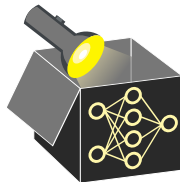


- Fit a linear model:

$$\hat{x}_2 = \hat{f}_{LM}(x_1) = \theta_0 + \theta_1 x_1$$

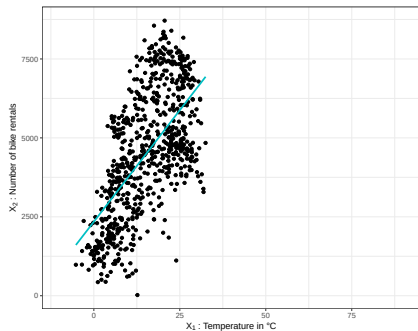
~> Slope $\theta_1 = 0 \Rightarrow$ no dependence

~> Large slope $\Rightarrow$ strong dependence

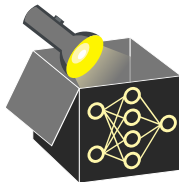


COEFFICIENT OF DETERMINATION R^2

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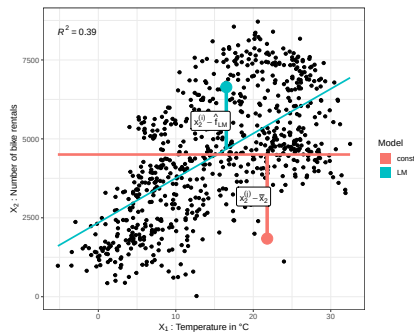


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 - ↪ Slope $\theta_1 = 0 \Rightarrow$ no dependence
 - ↪ Large slope $\Rightarrow$ strong dependence
- Exact θ_1 score problematic
 - ↪ Re-scaling of x_1 or x_2 changes θ_1
 - ↪ $^{\circ}\text{F} \rightarrow ^{\circ}\text{C} \Rightarrow \theta_1 = 78 \rightarrow \theta_1^* = 141$



COEFFICIENT OF DETERMINATION R^2

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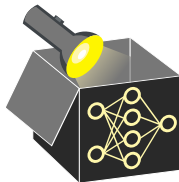
- Set SSE_{LM} in relation to SSE of a constant model $\hat{f}_c = \bar{x}_2$

$$SSE_{LM} = \sum_{i=1}^n (x_2^{(i)} - \hat{f}_{LM}(x_1^{(i)}))^2$$

$$SSE_c = \sum_{i=1}^n (x_2^{(i)} - \bar{x}_2)^2$$

$\Rightarrow$ Measure of fitting quality of LM: $R^2 = 1 - \frac{SSE_{LM}}{SSE_c} \in [0, 1]$

$\Rightarrow \rho(X_1, X_2) = R$



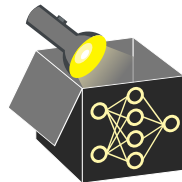
JOINT, MARGINAL AND CONDITIONAL DISTRIBUTION

For two discrete random variables X_1, X_2 :

Joint distribution

$$p_{X_1, X_2}(x_1, x_2) = \mathbb{P}(X_1 = x_1, X_2 = x_2)$$

p_{X_1, X_2}	$\mathbb{P}(X_2 = 0)$	$\mathbb{P}(X_2 = 1)$	p_{X_1}
$\mathbb{P}(X_1 = 0)$	0.2	0.3	0.5
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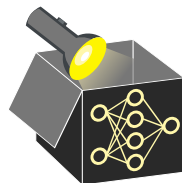
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Marginal distribution

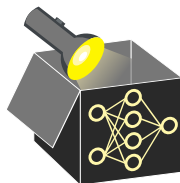
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Conditional distribution

$$\begin{aligned} p_{X_1|X_2}(x_1 | x_2) &= \mathbb{P}(X_1 = x_1 | X_2 = x_2) \\ &= \frac{p_{X_1, X_2}(x_1, x_2)}{p_{X_2}(x_2)} \end{aligned}$$

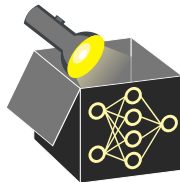
	$x_2 = 0$	$x_2 = 1$
$\mathbb{P}(X_1 = 0 X_2 = x_2)$	0.67	0.43
$\mathbb{P}(X_1 = 1 X_2 = x_2)$	0.33	0.57
Σ	1	1

DEPENDENCE

Dependence: Describes general dependence structure (e.g., non-lin. relationships)

- Definition: X_j, X_k independent $\Leftrightarrow$ joint distribution is product of marginals:

$$\mathbb{P}(X_j, X_k) = \mathbb{P}(X_j) \cdot \mathbb{P}(X_k)$$



DEPENDENCE

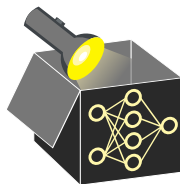
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$$\mathbb{P}(X_j|X_k) = \mathbb{P}(X_j) \text{ and } \mathbb{P}(X_k|X_j) = \mathbb{P}(X_k) \text{ (follows from cond. probability)}$$



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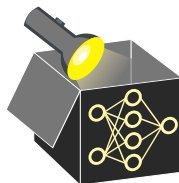
- Measuring complex dependencies is difficult but different measures exist

Examples

↪ Spearman correlation (measures monotonic dependencies via ranks)

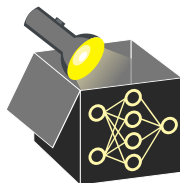
↪ Information-theoretical measures like mutual information

↪ Kernel-based measures like Hilbert-Schmidt Independence Criterion (HSIC)



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Examples

$\rightsquigarrow$ Spearman correlation (measures monotonic dependencies via ranks)

$\rightsquigarrow$ Information-theoretical measures like mutual information

$\rightsquigarrow$ Kernel-based measures like Hilbert-Schmidt Independence Criterion (HSIC)

- **N.B.:** X_j, X_k indep. $\Rightarrow \rho(X_j, X_k) = 0$ **but** $\rho(X_j, X_k) = 0 \nRightarrow X_j, X_k$ indep.

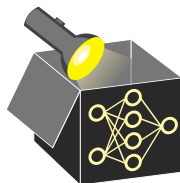
Equivalency holds if distribution is jointly normal

MUTUAL INFORMATION

- MI describes expected amount of information shared by two RVs:

$$MI(X_1, X_2) = \mathbb{E}_{p(x_1, x_2)} \left[\log \left(\frac{p(x_1, x_2)}{p(x_1)p(x_2)} \right) \right]$$

- MI measures amount of "dependence" between features by looking how different the joint distribution is from pure indep. $p(x_1, x_2) = p(x_1)p(x_2)$
 $\rightsquigarrow MI(X_1, X_2) = \mathbb{E}_{p(x_1, x_2)} \left[\log \left(\frac{p(x_1, x_2)}{p(x_1)p(x_2)} \right) \right] = \mathbb{E}_{p(x_1, x_2)} [\log(1)] = 0$
 $\rightsquigarrow MI(X_j, X_k) = 0$ if and only if the features are independent
- Unlike (Pearson) correlation, MI is also defined for categorical features



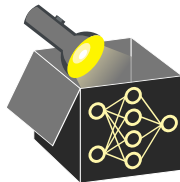
MUTUAL INFORMATION: EXAMPLE

For two discrete RV X_1 and Y :

$$MI(X_1; Y) = \mathbb{E}_{p(x_1, y)} \left[\log \left(\frac{p(x_1, y)}{p(x_1)p(y)} \right) \right] = \sum_{x_1 \in \mathcal{X}_1} \sum_{y \in \mathcal{Y}} p(x_1, y) \log \left(\frac{p(x_1, y)}{p(x_1)p(y)} \right)$$

X_1	...	Y
yes	...	yes
yes	...	no
no	...	yes
no	...	no

	$\mathbb{P}(X_1 = \text{yes})$	$\mathbb{P}(X_1 = \text{no})$	p_Y
$\mathbb{P}(Y = \text{yes})$	0.25	0.25	0.5
$\mathbb{P}(Y = \text{no})$	0.25	0.25	0.5
p_{X_1}	0.5	0.5	1



MUTUAL INFORMATION: EXAMPLE

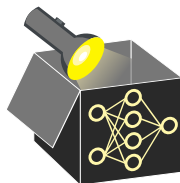
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X_1	...	Y
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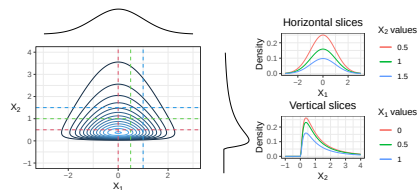
$$\begin{aligned} MI(X_1; Y) &= 0.25 \log \left(\frac{0.25}{0.5 \cdot 0.5} \right) + 0.25 \log \left(\frac{0.25}{0.5 \cdot 0.5} \right) \\ &\quad + 0.25 \log \left(\frac{0.25}{0.5 \cdot 0.5} \right) + 0.25 \log \left(\frac{0.25}{0.5 \cdot 0.5} \right) \\ &= 0.25 \log \left(\frac{0.25}{0.25} \right) \cdot 4 \\ &= 0.25 \log(1) \cdot 4 = 0 \end{aligned}$$



DEPENDENCE AND INDEPENDENCE

Example:

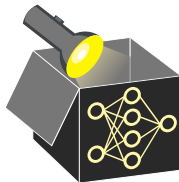
Independent



Conditional distributions at different vertical and horizontal slices (after normalizing area to 1) match their marginal distributions

$$\Rightarrow \mathbb{P}(X_1|X_2) = \mathbb{P}(X_1)$$

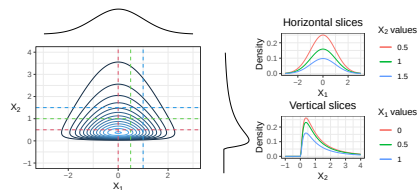
$$\mathbb{P}(X_2|X_1) = \mathbb{P}(X_2)$$



DEPENDENCE AND INDEPENDENCE

Example:

Independent

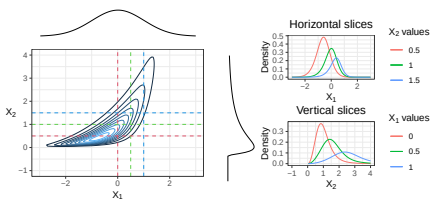


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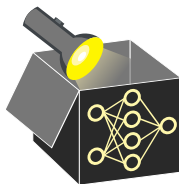
$$\Rightarrow \mathbb{P}(X_1|X_2) = \mathbb{P}(X_1)$$

$$\mathbb{P}(X_2|X_1) = \mathbb{P}(X_2)$$

Dependent



Conditional distributions do not match their marginal distributions

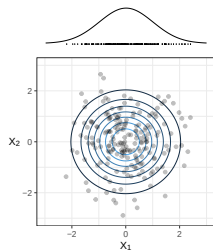


CORRELATION VS. DEPENDENCE

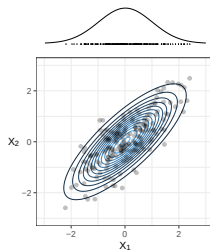
Illustration of bivariate normal distribution with different correlations

$X_1, X_2 \sim N(0, 1)$

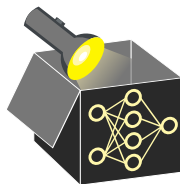
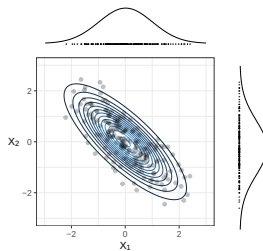
$\rho(X_1, X_2) = 0$
(independent)



$\rho(X_1, X_2) = 0.8$



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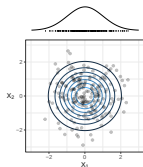


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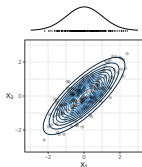
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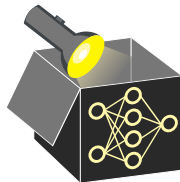
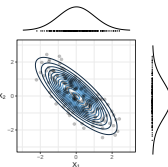
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Examples with Pearson's corr. $\rho \approx 0$ but non-linear dependencies ($MI \neq 0$):

$\rho(X_1, X_2) = 0$, $MI(X_1, X_2) = 0.52$ $\rho(X_1, X_2) = 0.01$, $MI(X_1, X_2) = 0.37$ $\rho(X_1, X_2) = -0.06$, $MI(X_1, X_2) = 0.61$

